# Final Review – Presentation

Mini Project or Internship Assessment (BCS 752)

- **Project Title: Real-Time Intraday Volatility Regime Classifier**


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**Session: 2026–27 (ODD SEM)**

# Introduction

## The problem

- 93% of individual F&O traders lost money in FY22–24; 70% of intraday cash traders lost money in FY23 (SEBI studies).
- Much of the damage happens in sudden volatility bursts that are hard to recognise while they are happening.
- India VIX is a 30-day measure; a price chart shows the burst only after the damage.

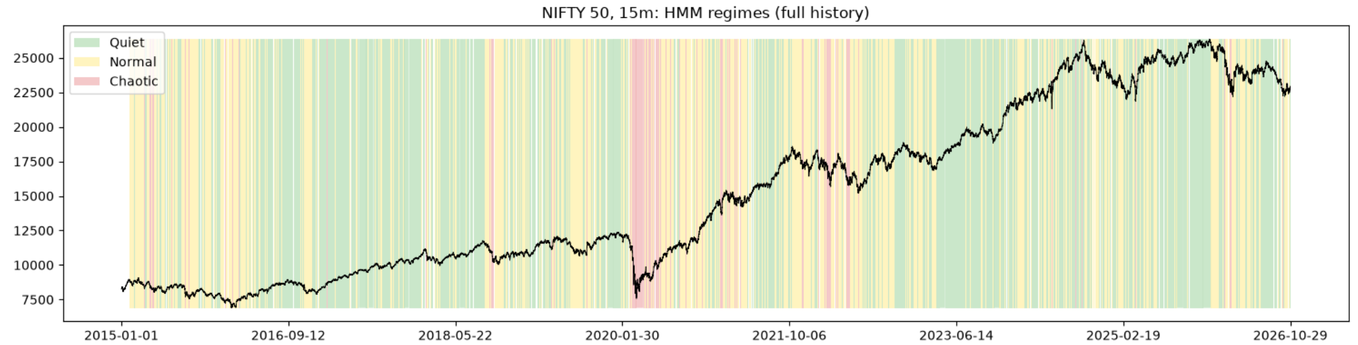
## Our answer

- VRDet classifies the NIFTY 50 every 15 minutes into a volatility regime and alerts on Chaotic. It nowcasts the current state; it does not forecast prices.

**Quiet**
Small, orderly moves · 49% of bars

**Normal**
Typical trading · 43% of bars

**Chaotic**
Violent moves, ~6× Quiet · 7.4% of bars, 342 episodes




*NIFTY 50 15-minute closes, 2015–2026, shaded by the hindsight regime (green Quiet, yellow Normal, red Chaotic)*

# Literature Survey

| Study | Method | Key finding |
|---|---|---|
| Hamilton (1989) | Markov-switching model | Market regimes as a hidden Markov state |
| Rabiner (1989) | HMM: forward filter, Viterbi | Real-time (filtered) vs hindsight (decoded) states |
| Andersen & Bollerslev (1997) | Intraday periodicity | Time-of-day pattern must be removed |
| Gu, Kelly & Xiu (2020) | 12 ML methods on stocks | Tree ensembles and shallow nets perform best |
| Grinsztajn et al. (2022) | Trees vs deep nets, 45 datasets | Trees still win on tabular data |
| Gunnarsson et al. (2024) | Review of ML volatility models | LSTM/GRU promising; GARCH/HAR still competitive |
| Kumar, Rao & Dhochak (2025) | VMD + ANN-LSTM-GRU (NIFTY, INFY) | Hybrid deep model best for volatility forecasts |

## Research gaps

- Most work forecasts a volatility number; few recognise the current regime in real time.
- Regime papers report hindsight states that use future data.
- Little intraday (15-minute) regime work on the NIFTY 50 with India VIX.
- Deep models often compared on one split, without confidence intervals.
- No free, local tool for retail traders.

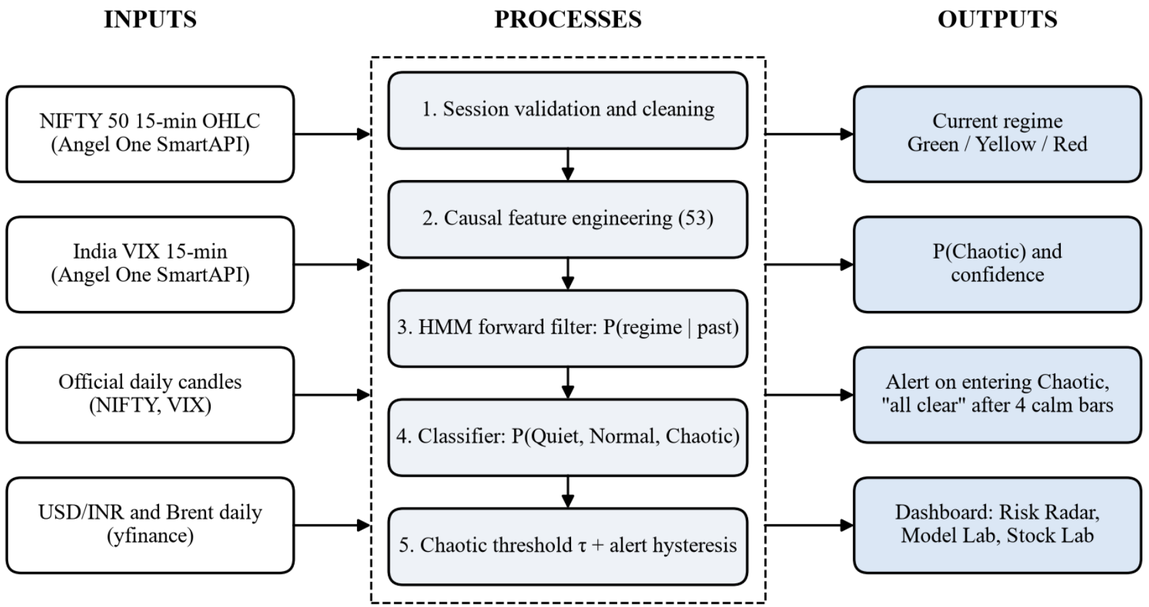
# Problem Formulation & Objectives

## Problem statement

Using only data up to bar t (NIFTY 50 and India VIX 15-minute candles, USD/INR and Brent daily), recover the regime that a hindsight HMM assigns to bar t, with high macro-F1, short delay and few false alarms, and show it to the trader within a minute of the bar closing.

## Objectives

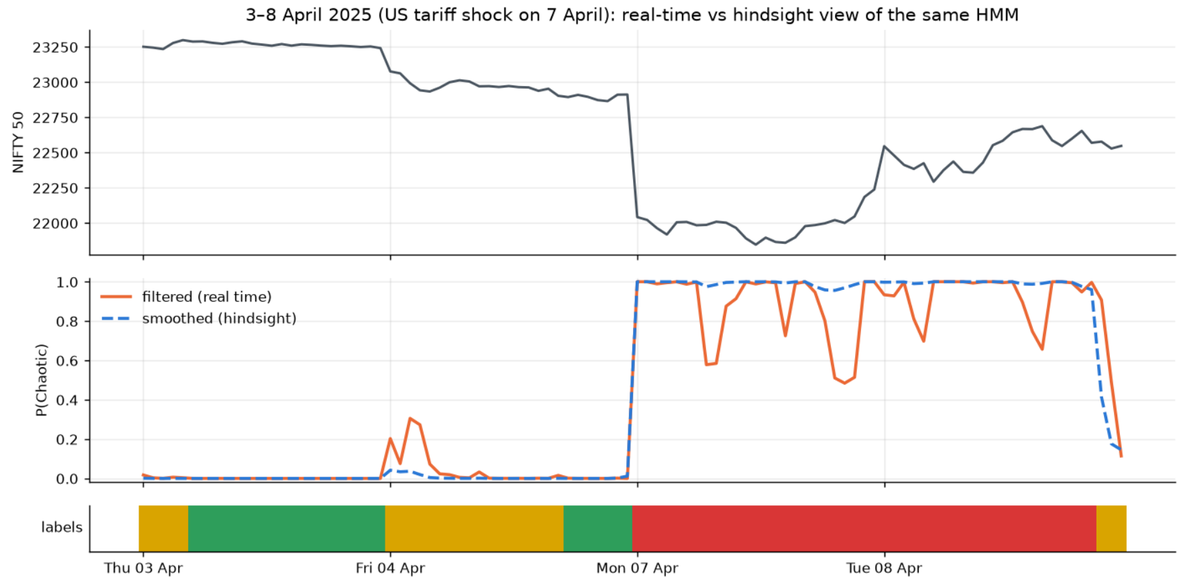
- Validated 11-year intraday archive from Angel One SmartAPI
- Credible hindsight labels (3-state HMM), checked on shock days
- Strictly causal features: bar t uses data $\leq$ t
- Beat the HMM-filter benchmark in a walk-forward test, 2022–26
- Compare 22 models with confidence intervals
- Local Streamlit dashboard on live data with alerts




*Inputs → processes → outputs*

# Methodology/ System Design/ Architecture

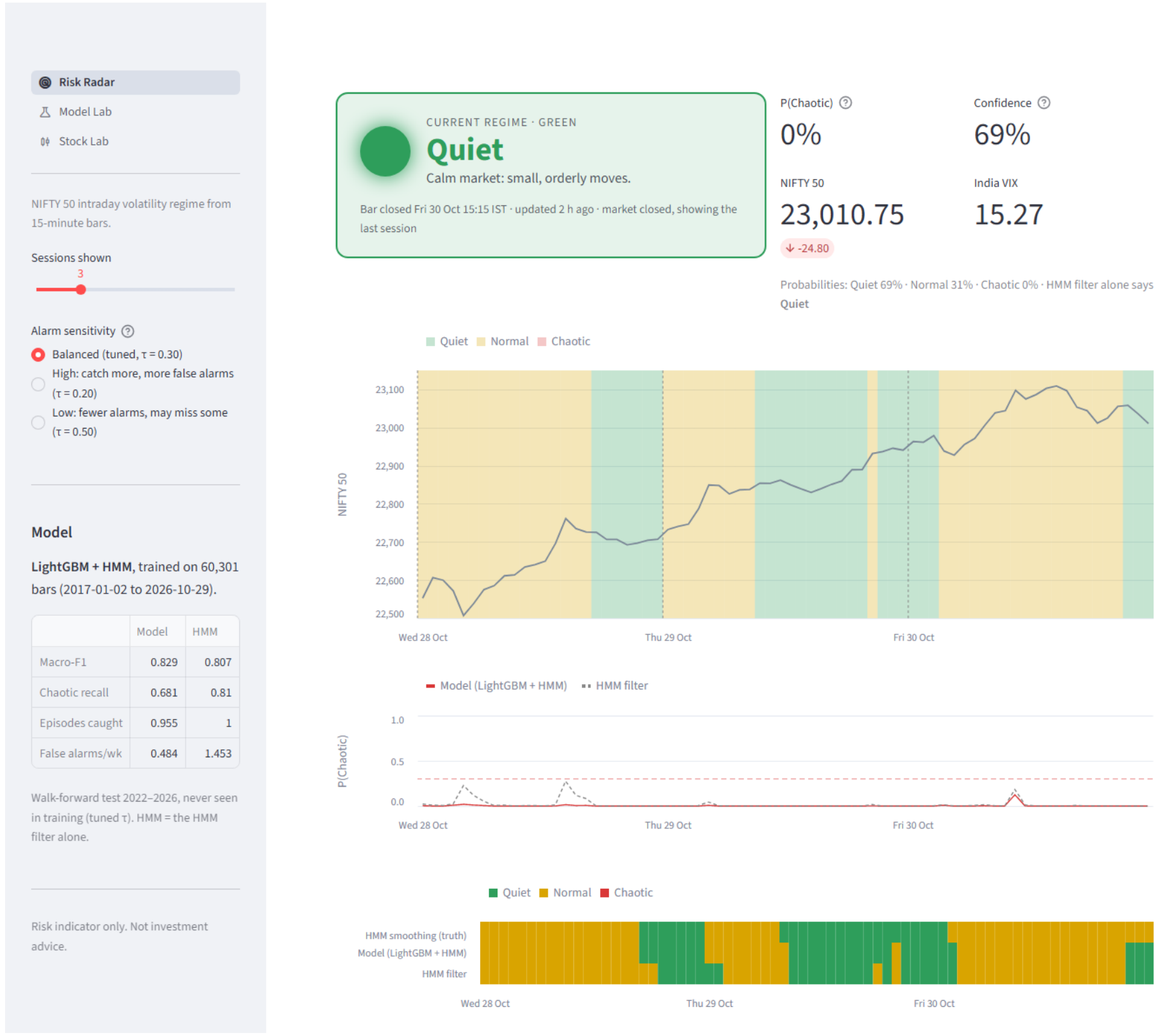
1. Download NIFTY 50 + India VIX 15-min candles (2015–26), USD/INR + Brent daily
2. Validate each session against official daily candles
3. Build 53 causal features (deseasonalised returns, range volatility, VIX, calendar, lags)
4. Label every bar: 3-state sticky HMM, Viterbi + 4-bar minimum (fitted 2015–21)
5. Add the HMM forward-filter probabilities → 56 inputs
6. Walk-forward: train on 2017..Y−1, tune τ on last year, test year Y (2022–26)
7. Score vs hindsight labels; bootstrap CIs; pick production model
8. Freeze model; live loop: fetch → features → score → alert → dashboard




*Same HMM, two views: real-time filter (wavers) vs hindsight smoother (decisive), 7 Apr 2025*

**Key idea:** truth = HMM in hindsight; benchmark = the same HMM in real time; the classifier must close the gap (filter agrees 85%, smoother 95%).

# Implementation




*Risk Radar: live regime, P(Chaotic), regime-shaded chart, truth row*

## Data

Angel One SmartAPI (TOTP login, chunked requests), yfinance; Parquet archive

## Models

Python 3.13, scikit-learn, LightGBM, XGBoost, CatBoost, PyTorch, hmmlearn

## Live runner

Every bar close + 45 s; one cycle ≈ 12 s; alerts with 4-bar hysteresis

## Dashboard

Streamlit, 127.0.0.1 only: Risk Radar, Model Lab (22 models, one-at-a-time queue), Stock Lab (~2,700 NSE stocks)

## Safety

Credentials in git-ignored .env; same feature code in training and live (verified: max difference 0.0)

# Results/ Discussion

**0.829**
macro-F1
HMM filter 0.807

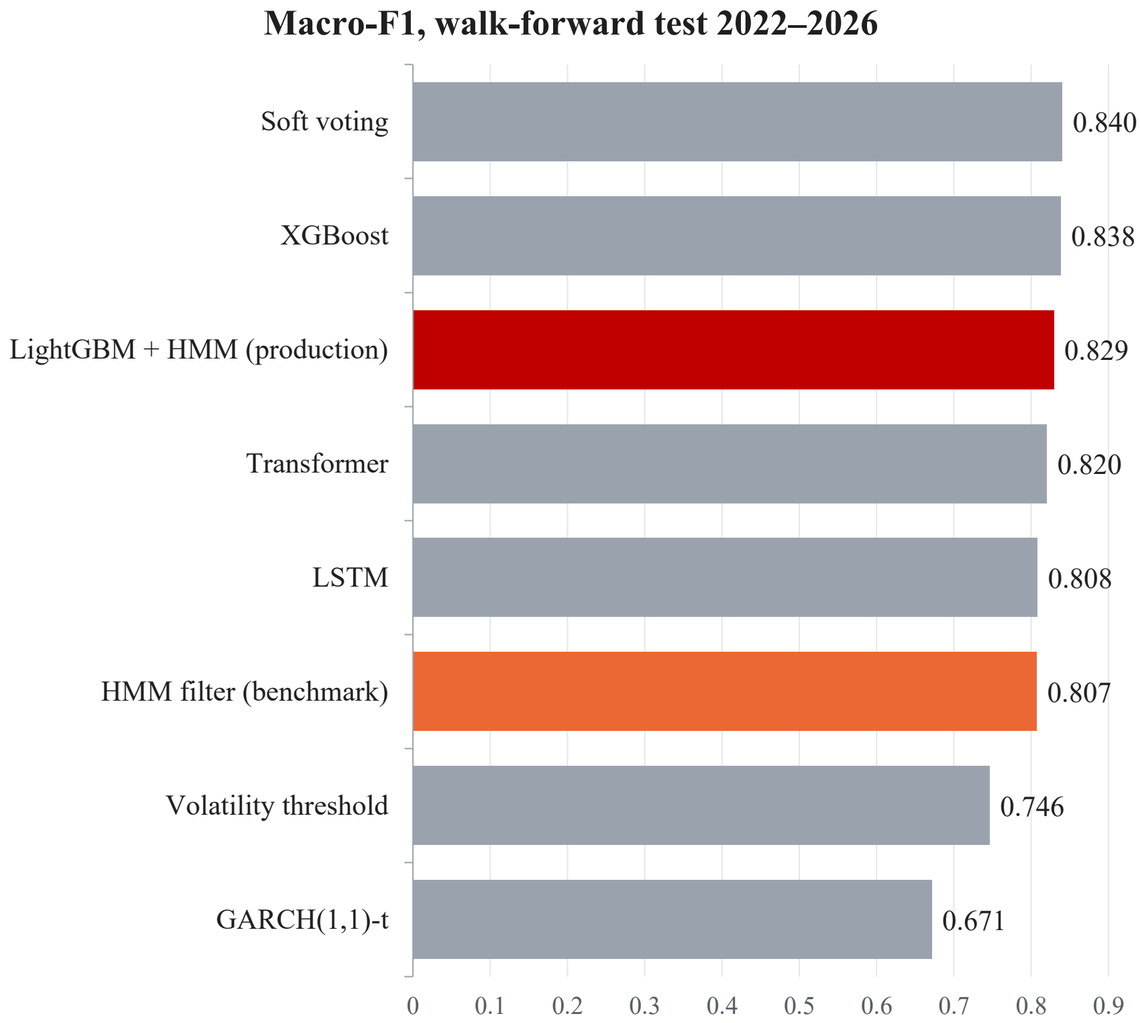
**0.48**
false alarms / week
HMM filter 1.45

**95.5%**
Chaotic episodes caught
134 test episodes

**1 bar**
median detection delay
15 minutes

## Discussion

- Production LightGBM + HMM beats the HMM filter in every test year and cuts false alarms by two thirds.
- Of 22 models, 20 beat the benchmark; only soft voting and XGBoost are significantly better (+0.01). Most are statistically tied.
- LSTM and GRU are significantly worse; learning curves show the limit is the number of Chaotic episodes, not bars.



# Conclusion/ Future Work

## Conclusion

- Validated 11-year NIFTY 50 + India VIX intraday dataset; vendor defects found and fixed
- Credible hindsight regimes: all 10 major shock days flagged Chaotic
- Real-time classifier beats the honest benchmark: 0.829 vs 0.807 macro-F1, 0.48 vs 1.45 false alarms/week
- Small, well-validated models are enough; deep models are not better with this much data
- Working local app: Risk Radar, alerts, Model Lab, Stock Lab

## Limitations

- Truth is itself a model (HMM); other labelers disagree ($\kappa \approx 0.5$)
- Few episodes (134 in test) make small differences unresolvable
- Detects chaos once it starts (1-bar delay); cannot warn before the first violent bar
- Single-broker data dependence; stocks are harder than the index

## Future work

- Early warning model for the next bars, with event calendars
- Futures OI, option skew, breadth and news sentiment as inputs
- Embargoed validation, calibration and input-group ablations
- Pooled training across stocks to get more Chaotic episodes
- Soft-voting ensemble in production; secure multi-user deployment

# References


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# *Thank You*